Name: Gina Nguyen

Personal Programming Project Report

Title: Campusly

A student productivity app created for college students made by a student.

1. **(5pts) Honor Code and LLM Usage for this Report.**
   As a Hokie, I will conduct myself with honor and integrity at all times. I will not lie, cheat, or steal, nor will I accept the actions of those who do.

   During the preparation of this assignment, Gina Nguyen used Claude.ai to help program my project. After using this tool, I/we reviewed and edited the content as needed to ensure its accuracy and take full responsibility for the content in relation to grading.

2. **(15pts) Learning Objectives:**
   - Learning Objective 1 (conceptual understanding): Learning how to implement multiple academic tools into one efficient application using software design and data organization practices learned in class
   - Learning Objective 2 (skill development): Developing proficiency in Python, HTML, and CSS when building an interactive webpage

I think I did meet those objectives because I learnt to combine tools into one webpage using the software design practices we learnt in class. For instance, I learned how to brainstorm on tackling both parts of a program, including the front-end design with HTML and CSS, and the backend using Python. For skill development, I definitely did develop a proficiency in Python, HTML, and CSS when building the interactive webpage by visiting tutorial pages and asking an LLM questions about the logic and how each piece of code works. I also learnt something new using Flask to connect the frontend and backend. It also taught me troubleshooting skills because I had issues with running the server and getting it to work properly.

3. **(15pts) Timeline:**
   Outline how you spent time on your project. Break down the time into specific tasks or milestones. Here is an adjustable schedule to get you started. Actual Details should be 50-100 words each and should compare or reflect on differences from your proposal.

Time	**Task**	**Expected Details from Proposal**	**Actual Details**
Hour 1-2	Research and gather resources	I'll be researching various student	I did note significant features that would be useful to add to my web page. I also found

Time	Task	Expected Details from Proposal	Actual Details
		productivity apps, course planners, and GPA calculators to note useful features and get inspiration for interface design. This will also help me gather ideas for organizing various academic tools into a single application. I will also review tutorials for Python, HTML, and CSS to help me with actually programming it.	inspiration for my front-end design specifically from Notion. I also learned more about Flask to connect my front-end design to the back-end design when implementing my project in Python. I used an LLM to help me learn, and with tutorials and classwork on what we learnt with HTML and CSS, I was able to program the front end in one style sheet.
Hour 3-4	Design the project structure and plan	For the design, I'll create and sketch out a layout for the application, including the GPA calculator, course organization, and to-do list. I'll also plan how they'll navigate through the app and what features will be essential for the first version. I'll also seek feedback from peers to see if I can create the best design I can.	After noting features I want from webpages I'm taking inspiration from, I drew out a sketch of how I wanted the layout to be. I also decided to create tabs to include special features such as a to-do list, course organization, and a GPA calculator. This will make it easier for the user to navigate. I asked my friends to
Hour 5-8	Start coding the basic functionalities	I'm going to start on the main structure using Python, HTML, and CSS. I'll be working on the home page layout first while prioritizing functionality. I'll then work on how it looks after I'm content with	I did focus on building the user interface first using HTML and CSS. I first created the skeleton of the webpage using HTML with assistance from an LLM and tutorials from the course page. After I was content with the layout in HTML, I then styled it using CSS. I spent more time than expected

Time	Task	Expected Details from Proposal	Actual Details
		how it works, gradually testing its functionality. I'll be including a working calculator, organizer, and to-do list.	working on HTML and CSS because I had to figure it out along the way. For the backend with Flask and Python, with LLM assistance, I also created routes to add courses, tasks, and GPA entries. This was the most challenging part because I had to figure out how to make a functioning calculator while connecting it back to my front-end design. I also had help from the LLM to figure out the logic.
Hour 8-11	Test and debug the initial version	I'll be testing each feature of the application to ensure everything functions correctly. This includes test cases ot check that the GPA calculator gives accurate results, tasks being added and removed properly. I will fix errors, issues with user input, and also layout issues. I'll also improve the organization of the app so that users have an easier time using and navigating.	I needed an additional hour to debug and test my webpage because I did have some trouble running it. With the help of an LLM, I was able to properly test and open up my webpage. This took way longer than expected because I ran into problems with my Python file not running, routes taking me to the wrong thing, as well as design and layout issues, because I was not happy with what it looked like. In the end, I was able to fix it until I was happy with it.
Hour 12	Refine and add advanced features	I'll continue to ask for feedback to improve the visual design. This includes adjustments to the spacing, colors, and layout. I also plan to add features such as task priority labels, due date reminders, and a	I did add more advanced features after fixing up my webpage. I asked my peers to help me improve my design and added priority tags after my to-do list was fully functional. I then also added a dashboard with different cards to summarize user GPA, course

Time	Task	Expected Details from Proposal	Actual Details
		progress tracker for different course assignments. I may even add a dashboard to summarize upcoming deadlines, tasks, and even grades.	number, and remaining tasks left to do.

4. **(55pts) Final Product Description:**
   Include your proposed MVP, Target, and Reach versions.
   i. Minimum Viable Product (MVP): A web application that combines a GPA calculator, course organizer, and task list. The target audience is college students looking for a quick way to access all academic tools in one place, instead of using multiple apps. Built with Python, HTML, and CSS.

   ii. Target Product: A student productivity app that includes a GPA calculator, course organizer, assignment due date reminders, and a task completion dashboard. To-do lists will include additional features, such as high-priority labels. Built using Python, HTML, and CSS. The target audience is anyone who is looking for more features in tracking their assignment progress and plans to return to the app.

   iii. Reach Version: An advanced version of the app with more added features, including login accounts, different themes such as dark mode and light mode, reminders, grade predictions, and even a mobile-friendly design. Students can plan semesters and even track long-term academic progress. The target audience is college students looking for an advanced academic planner with personalized tools to use long-term.

   iv. (20pts) Description of final product including target audience, user story, problem statement, key features, technical details and technologies used. (100 – 150 words):

   College students often struggle with managing their assignments and courses and have no way to organize them in one place. They usually have to resort to multiple platforms, including a GPA calculator, planners, notes apps, or reminder apps. My personal project, Campusly, provides a solution to this by having it all in one place. The target audience is college students who want a simple and efficient way to manage their academics. As a student, I want an efficient way to manage everything at once to reduce stress. Key features of Campusly include a to-do list with priority tabs, a course organizer, and a GPA calculator with an overview dashboard. This product was built using Flask to connect the HTML and CSS frontend with the Python backend.

v. (20pts) Provide a YouTube link to your video demonstration (1–2 minutes, narrated). **Link**

vi. (15pts) Any input files, coding files, and test files should be uploaded. Provide a list here of file names and purposes, or any links to live sites or artifacts. Remember, code should also be commented. A README file should be created and uploaded so that we have the option to follow your instructions to run your project.

Project Repository & Code Submission Details:
Project Repository on git.cs.vt.edu (other options are code.vt: https://code.vt.edu/ or (Git-hub only if you are part of virtual global collaboration)): Your repository should be well-organized, documented, and easy to navigate. At a minimum, include the following structure:

- **code/** – All source code files for your project (organized by component or module if applicable).
- **data/** – Any input files, datasets, or configuration files used by your program.
- **tests/** – Test scripts or files demonstrating how your code was verified.
- **docs/** – Supporting materials such as screenshots, reports, or documentation.
- **report/** – This final report document.
- **README.md** – A detailed file describing:
    - Project overview and purpose
    - Video link of your project
    - Installation and setup instructions
    - How to run the program and reproduce results
    - Technologies or libraries used
    - Author(s) and contribution summary

Required:

- Maintain a logical directory structure, do not store all files at the root level.
- Include comments in your code to explain logic and design decisions.
- Keep your repository **private** until grades are released, then you may make it public.

Share access with the following personnel (Add them as collaborators):

GTA Name	Section(s)	Professor
Mona Moghadampanah	13392 & 13394	S. Cao
Katelyn Crumpacker	13393 & 13395	M. Ellis

5. **(10pts) Consultation and Use of LLMs:**
   Each student must create a unique project but is allowed to consult with other people and use Large Language Models (LLMs). Describe how you incorporated these resources into your project:

- **Consultation Description:**
  I asked my classmates in my CS class or my friends to look at my webpage and note things they liked about it and what they didn't. I then used that information to improve my design, and then checked up with them again.
- **Use of LLMs:**
  If I didn't understand logic or how to do something with the code, I would prompt to help me figure it out. It helped me find ways to test my code and then debug. If I was confused with learning how to use Flask, I also prompted to help walk me through it and use the functions that are imported from that. It helped me generate ideas as well when I did not know how to convert an idea into an actual program. I also made sure to fully understand every piece of code written.